

Twenty topics, four learning regimes, one question: *how much representation matters.*

What is each step up the representational hierarchy actually worth?

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Twenty classes, several of them near-duplicates
(`comp.sys.ibm.pc.hardware` vs `comp.sys.mac.hardware`;
`soc.religion.christian` vs

`talk.religion.misc`). We use 20 Newsgroups as a controlled testbed — sparse statistics, dense embeddings, pretrained transformers, billion-parameter LLMs — and measure what each rung up the ladder is worth.

Q1

Shallow MLP

word2vec · TF-IDF ·
mean+max pool

Q2

BERT fine-tune

end-to-end transformer

Q3

MLM → fine-tune

domain-adaptive pretraining

Q4

Llama-3 prompting

zero / few-shot, frozen

+

QLoRA bonus

parameter-efficient LLM tuning

18 846

DOCS · 20 CLASSES

5

METHODS COMPARED

79.9%

BEST MACRO-F1

20 Newsgroups, *stripped of leakage*.

Headers, footers and quoted replies leak the label. Strip them out and the baseline falls from tutorial-grade 90 % into the low 60s.

*f*etch_20newsgroups(remove=("headers", "footers", "quotes")). Roughly class-balanced (377–600 docs/class), heavy-tailed length: median ~40 tokens, p95 ~280, max above 6 700.

TRAIN DOCS

11 314

stratified

TEST DOCS

7 532

held-out by-date split

CLASSES

20

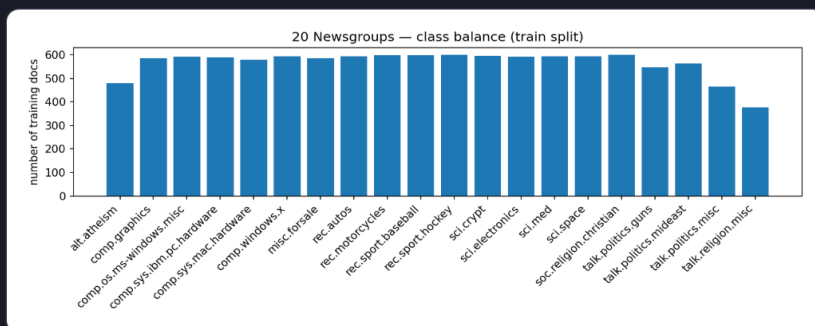
topic categories

RAW VOCAB

73 118

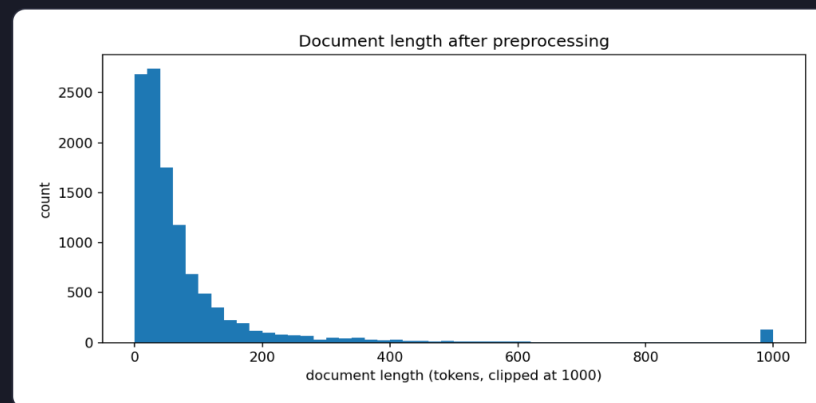
after preprocessing

Class balance



Train-set document counts per class. Mild imbalance – talk.religion.misc and talk.politics.misc are the smallest classes and consistently the hardest to recover.

Document length distribution



Long tail informs the BERT max_length=256 choice – covers ≥95 % of documents without paying for tokens at the extreme.

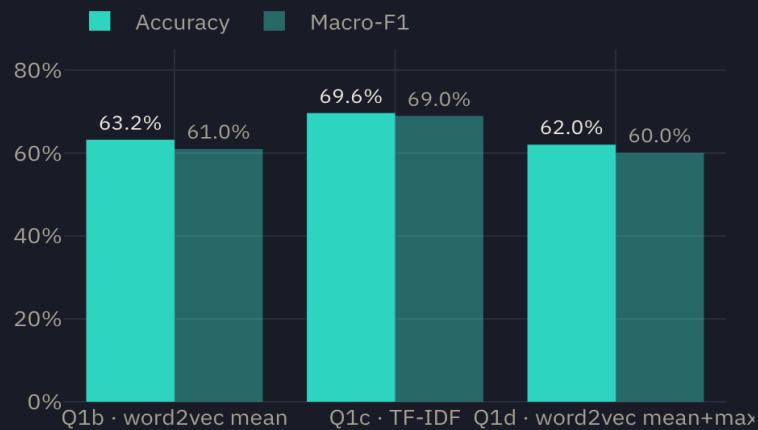
Why this matters. Stripped-text baselines start at ~63 % rather than the inflated 90 %+ in tutorial code — so every Q1→Q2→Q3 lift is genuine, not spurious.

Shallow MLP, *three representations*.

One classifier, three inputs. Any accuracy difference is the representation.

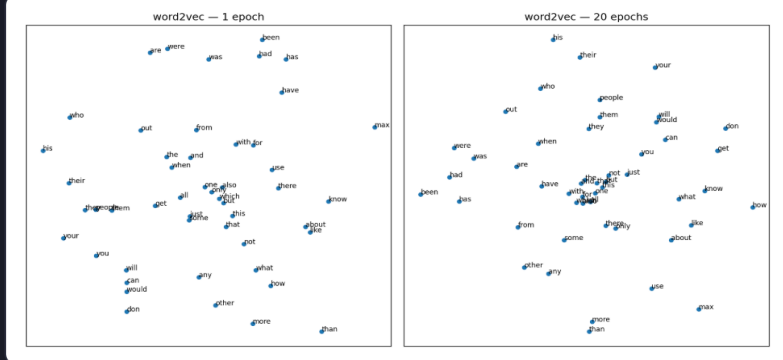
A 2-layer MLP (input $\rightarrow 256 \rightarrow 20$), held fixed across word2vec mean-pool, TF-IDF, and mean+max pooled word2vec. The classifier never moves; only the input does.

Accuracy & macro-F1 by representation



TF-IDF wins the group. Adding max-pool to mean-pool hurts: the max dimension is dominated by high-magnitude function tokens.

Trained vs untrained embeddings



t-SNE of 500 tokens at 1 epoch (left) vs 20 (right). Vocab is identical (19 306); topic clusters only emerge with training — the lift is the embedding, not the lexicon.

Design decisions

- **Stopwords kept for word2vec, dropped for TF-IDF.** Word2vec needs function words to stabilise the context window; TF-IDF doesn't.
- **Determinism via workers=1 + seed=42.** Gensim's multi-threaded training is non-deterministic — the 1-vs-20-epoch comparison would otherwise be meaningless.
- **Mean+max pool as a no-architecture-change probe.** The spec required improving Q1 without touching the network. Doubling the input dim with max-pool isolates representation from capacity.

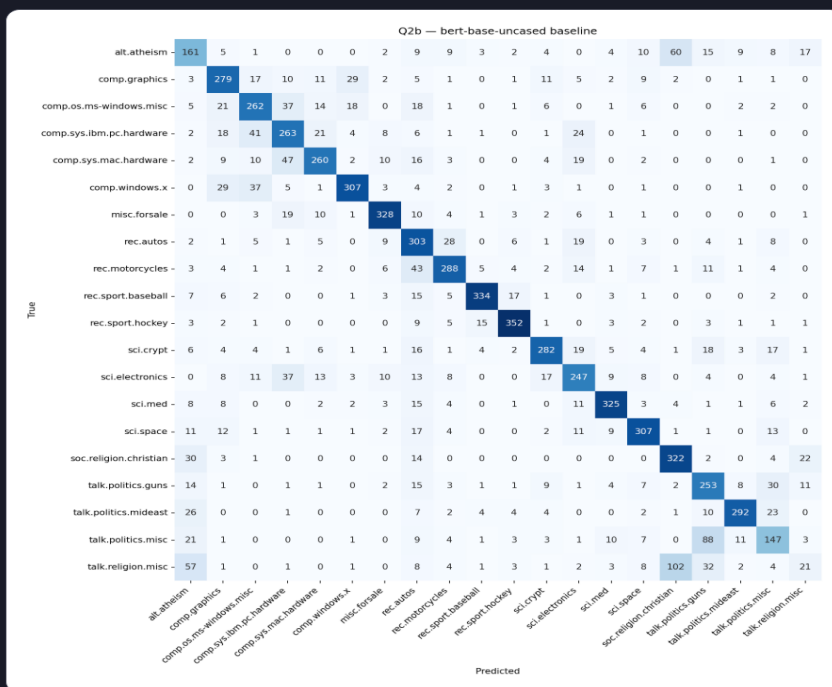
BERT-base, *fine-tuned end-to-end*.

110 M pretrained parameters, same val indices as Q1. The lift is real because the split is identical.

End-to-end fine-tune of bert-base-uncased via HuggingFace Trainer. Same seed=42 as every other stage — apples to apples.

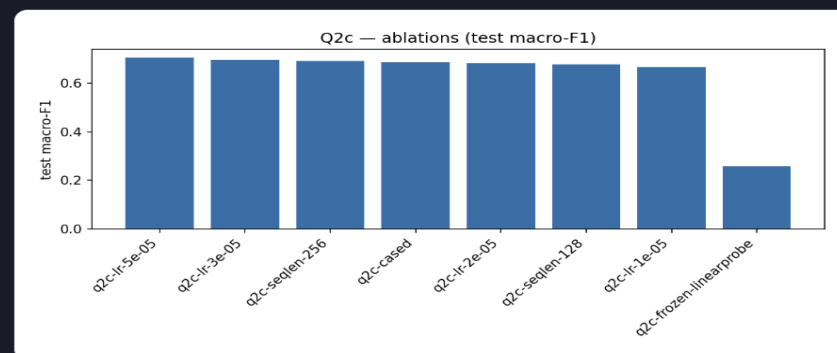
Q1 BEST (TF-IDF) **69.6%** → Q2C (BEST LR) **71.9%** +2.2 pts

Q2b · baseline confusion



Diagonal-heavy. Residual error concentrates inside the comp.*, politics, and religion sibling clusters.

Q2c · learning-rate sweep



Best LR: 5e-05 — the canonical BERT fine-tuning rate. Larger destabilises pretrained weights; smaller under-trains in three epochs.

Design decisions

- **max_length=256.** ~95th percentile of token-length — covers nearly every document without paying for outliers.
- **Index parity with Q1 and Q3.** One data.train_val_split(seed=42) call, reused everywhere — same val set throughout.
- **MPS-safe Trainer flags.** fp16=bf16=pin_memory=False; load-best on macro-F1, not loss.

MLM pretraining, *then fine-tune*.

Two stages; Q2b's hyperparameters held verbatim in stage B. The only variable is the encoder init.

Continue MLM on the 20NG train corpus, then fine-tune with Q2b's hyperparameters held verbatim. Anything that moves is the MLM stage.

MLM PERPLEXITY

7.71

held-out MLM split

CLASSIFICATION ACC

71.6%

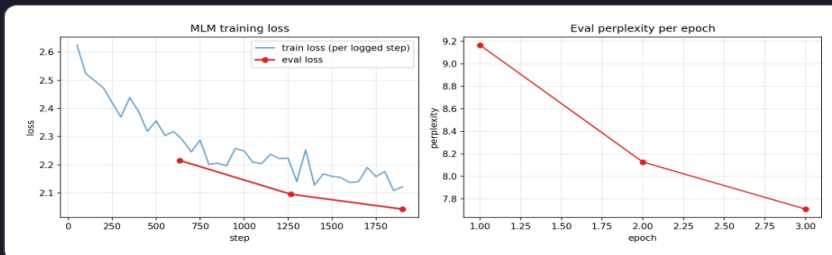
after MLM → fine-tune

Δ VS Q2B

0.80 pts

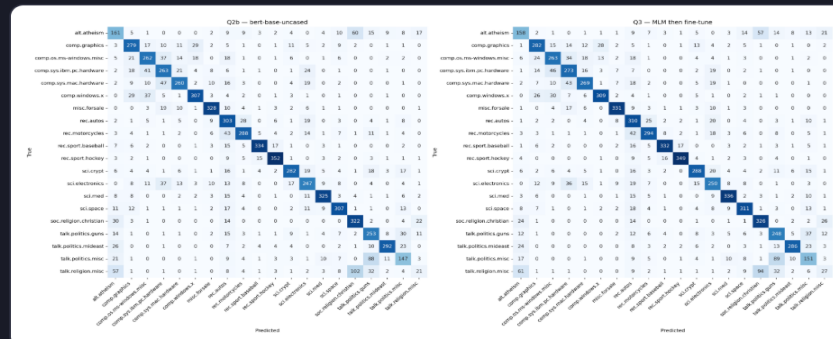
marginal lift

Stage A • MLM loss



Eval loss ~2.04, perplexity ≈ 7.71. The MLM-eval 90/10 split is internal to the train corpus; the classification val set is never touched.

Stage B • classification confusion (Q2b vs Q3)



Visually almost indistinguishable. DAPT is real but marginal here — the corpus is too small and too close to the English BERT already knows.

Design decisions

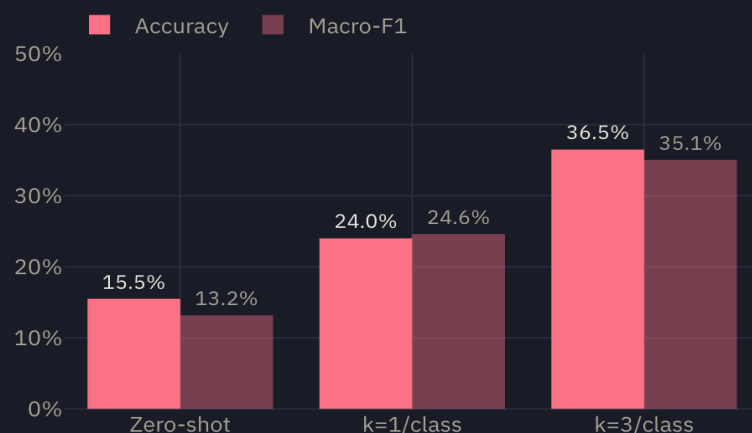
- **Independent MLM-eval split.** 90/10 carve *inside* the training docs — the classification val set is held out from MLM exposure.
- **15 % masking rate, untuned.** BERT default. The question was "does DAPT help *at all*", not "what's the optimal mask rate".
- **Stage B = Q2b hyperparameters, verbatim.** Only the encoder init changes.

Llama-3, *zero-shot & few-shot*.

3 B parameters, frozen, 4-bit. Classification by generated text — parsing is its own diagnostic.

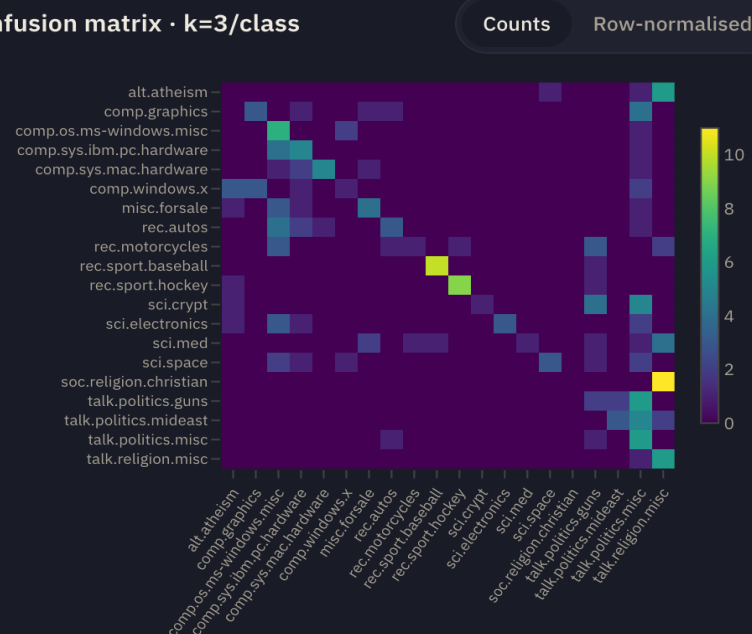
Frozen meta-llama/LLama-3.2-3B-Instruct in 4-bit nf4, prompted via chat template. No gradients; output parsed back to a label with substring + Levenshtein fallback.

Demonstrations vs accuracy



$k=0 \rightarrow k=1 \rightarrow k=3$. Few-shot more than doubles zero-shot; even $k=3$ (60 demos in context) is well below fine-tuned BERT.

Confusion matrix · k=3/class



Invalid-parse rate: zero 0.5% · k=1 4.0% · k=3 3.5%. The diagnostic separating instruction-following failures from genuine misclassifications.

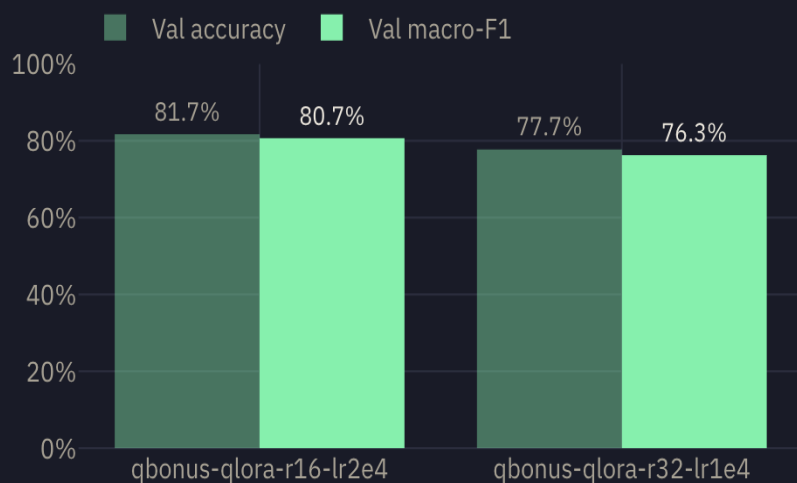
Honesty note. Q4 evaluates on a stratified 200-doc subset (10/class, seed 42). The full 7 532-doc test set would have been ~6 h per run. The same subset is reused for Q-bonus to keep the comparison like-for-like.

QLoRA fine-tune *of Llama-3*.

Same base as Q4 – this time, allowed to train.

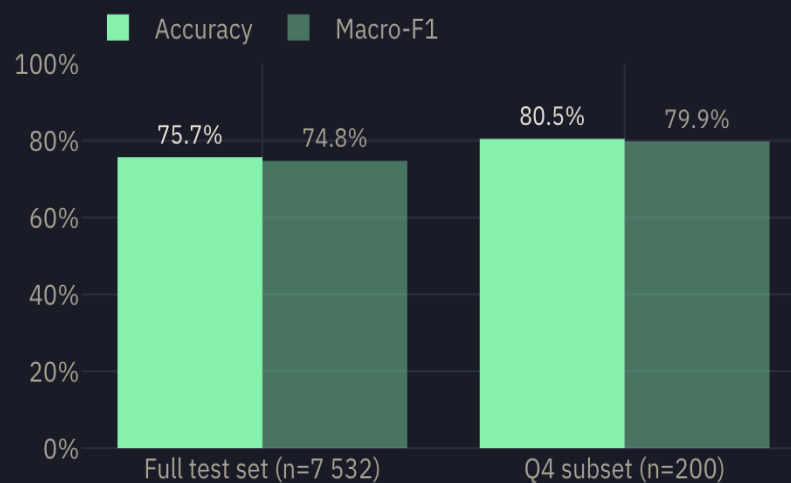
LoRA on {q,k,v,o}_proj; classification head score trained in full precision via `modules_to_save`. A two-run sweep over rank and LR picks the winner by val macro-F1.

Sweep · validation macro-F1



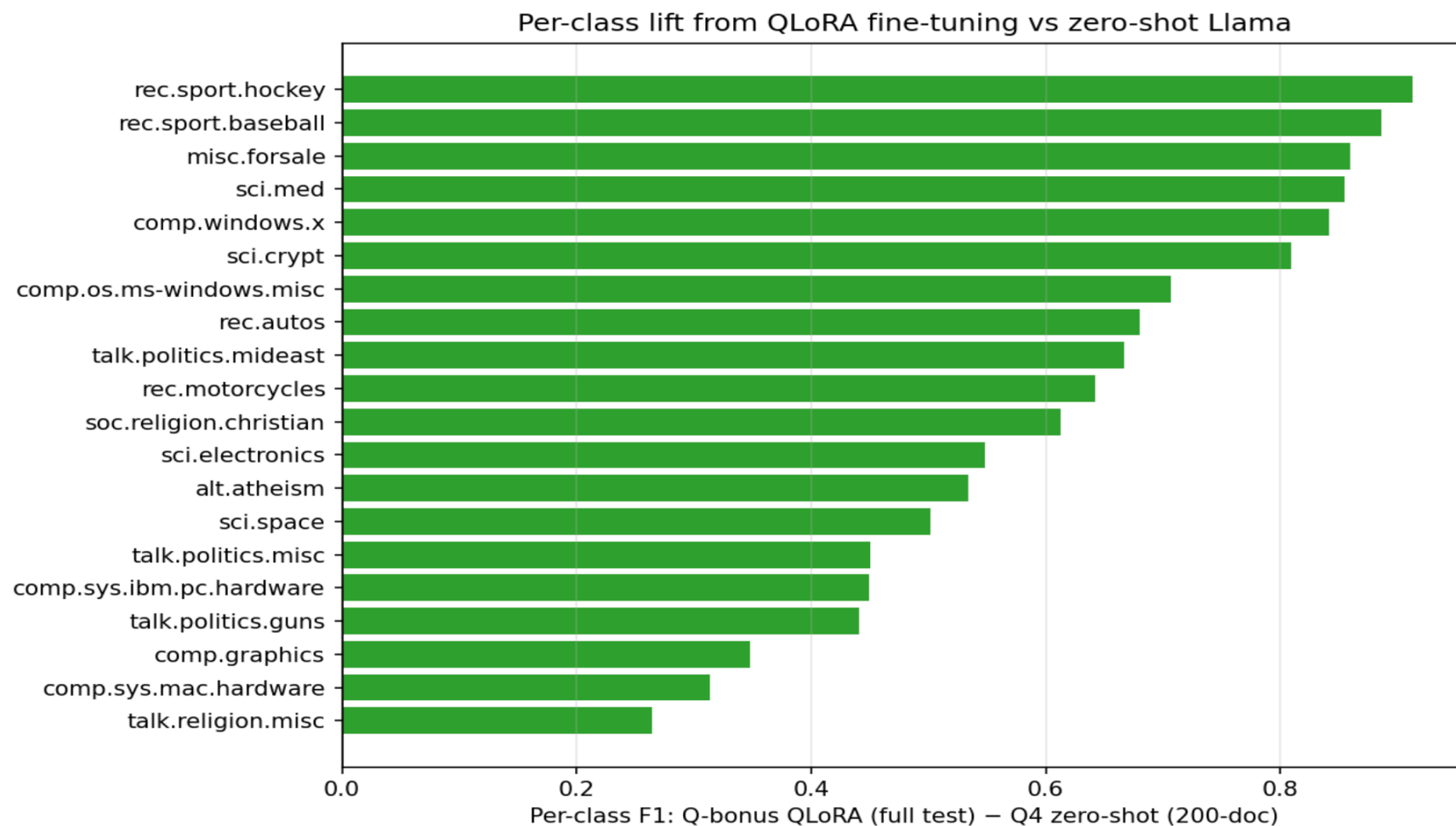
Winner: qbonus-qlora-r16-lr2e4. Rank 16/LR 2e-4 edges rank 32/LR 1e-4 – adapter capacity isn't the bottleneck here.

Winner · final test metrics



Evaluated twice: full 7532-doc test set (vs Q1/Q2/Q3) and the 200-doc subset (byte-identical to Q4).

Per-class delta vs Q4 k=3



Per-class F1 lift on the 200-doc subset. Sibling classes (computer-hardware, religion, politics families) gain the most – exactly where prompting struggled.

Design decisions

- **LoRA on attention projections only.** Standard and well-validated locus — rank 16 / 32 specialises attention without retraining 3 B parameters.
- **modules_to_save=['score'].** If the freshly-initialised head is wrapped by LoRA, it cannot move enough to learn — loss looks fine, model never classifies. Saving in full precision is the fix.
- **Two evaluations, one winner.** Full test → vs Q1/Q2/Q3. Q4 subset → vs zero/few-shot.

All five methods, *on one axis*.

One chart, the categorical palette used throughout. Colour for the question family; height for the finding.

The headline arc, in one frame.



What the chart says. Representation matters more than scale. Q1c → Q2 is large; Q3 is flat (corpus too small for DAPT). A frozen 3 B LLM is *worse* than fine-tuned BERT; once it's allowed to train via QLoRA, it overtakes everything — but by far less than parameter count would suggest.

Lessons *learnt*.

Six findings, one consistent surprise. The places parameter count was supposed to substitute for engineering — it didn't.

*Seven experiments, **one consistent surprise**. Parameter count never substituted for engineering; engineering details I'd dismissed as bookkeeping kept deciding the result.*

01 · The headline

Representation, *not scale*, did the work.

110 M parameters of pretrained BERT beat a frozen 3 B Llama-3 by ~35 points on the matched subset. Even QLoRA'd, the 3 B model only edges BERT by ~4 points. Task-specific gradient flow beat both context length and parameter count.

● Q2 · BERT

● Q4 · Llama prompted

● Q-bonus · QLoRA

02

In-domain MLM is *not* a free lunch.

Continued MLM lifted Q2b → Q3 by less than one point — well inside seed noise. 11 k documents is too small and too close to generic English. The paper-claim holds at a different order of magnitude.

● Q3

03

Classification heads *fail quietly*.

QLoRA without `modules_to_save=[ 'score' ]` trains, logs sensible loss, and never classifies. The framework gives no warning. Defaults are not safe defaults for novel head wiring.

● Q-bonus

04

Free wins exist *if you look*.

Concatenating max-pool to mean-pool doubled the input with zero new parameters. Engineering can substitute for capacity when the bottleneck is representation, not learning.

● Q1d

05

Identical splits, *or nothing*.

Q1 → Q2 → Q3 lifts mean what they claim only because every stage reuses `train_val_split(seed=42)`. Without that contract, “Q2 beats Q1” is indistinguishable from “Q2 got luckier.”

● Q1

● Q2

● Q3

06

Disclosure beats *laundering*.

The Q4 subset is ~5 points easier than the full test set — visible in the gap between Q-bonus full vs subset of the same adapter. Stating it up front protects the comparison.

● Q4

● Q-bonus

Future *work*.

Four leads, four follow-up experiments. Each one tests a hypothesis raised but not settled by the work in this issue.

Full-test Llama evaluation

Re-run Q4 and the Q-bonus winner over the 7 532-doc test set.
Removes the lingering uncertainty around the 200-doc subset.

Larger MLM corpus for Q3

Concatenate 20NG with an external news / forum corpus before the MLM stage. Does the DAPT lift appear once the LM sees new vocabulary?

Instruction-tuned Llama-3.1-8B

The 3 B model already comes within ~5 points of fine-tuned BERT once QLoRA'd. Same training budget, larger model — does the gap close?

Error analysis on sibling classes

Every method's residual error concentrates in the same three clusters (comp . *, politics, religion). A focused study would clarify whether the failure is representational or label-overlap.